ثانوية المدينة التعليمية
Education City High School

2.2

Change in Linear and Exponential Functions

Core-to-Challenge Worksheet and AP Practice

AP Precalculus - Unit 2 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Student Information

Name: ____________________ Class: __________ Date: __________

Directions

Show mathematical evidence. Retain domain restrictions, label graphs, include units, and write complete interpretations. Calculator use is permitted only when the item explicitly requires approximation, regression, or numerical solving.

Readiness

1. Write one precise sentence explaining this principle: Linear functions add a constant amount over equal intervals.

2. Write one precise sentence explaining this principle: Exponential functions multiply by a constant factor over equal intervals.

3. Write one precise sentence explaining this principle: A percent rate p corresponds to factor $1+p$ for growth and $1-p$ for decay.

4. Write one precise sentence explaining this principle: Two points determine either model once the family is specified.

Core

5. Construct a linear function with $L(2) = 11$ and $L(7) = 31$.

6. Construct an exponential function with $E(0) = 6$ and $E(3) = 48$.

7. Compare a 5% yearly increase with adding 5 units yearly.

8. State what the formula $L(x) = L_0 + mx$ tells you and name any required restriction.

9. State what the formula $E(x) = E_0 b^x$ tells you and name any required restriction.

10. State what the formula $b = \left(\frac{E(x_2)}{E(x_1)}\right)^{1/(x_2 - x_1)}$ tells you and name any required restriction.

Medium

11. Construct the linear function through (0, 5) and (4, 17).

12. Construct the exponential function through (0, 5) and (4, 80).

13. Construct the linear function through (2, 10) and (7, 35).

14. Construct the exponential function through (1, 6) and (3, 54).

15. Construct the linear function through (0, 12) and (5, 6).

Hard

16. A value increases from 40 to 52 in 3 years. Compare the linear yearly change and exponential yearly factor.

17. Find when $10 + 4t = 10(1.1)^t$ first holds after t=0 approximately.

18. Explain why an exponential with base greater than 1 eventually exceeds any increasing linear function.

19. A table has first differences 3,4.5,6.75 and ratios 2,1.75,1.64. Is linear or exponential exact?

Difficult

20. Find all x where $2x + 8 = 8 \cdot 2^{x/4}$.

21. An exponential and line are tangent-like at x=0 in the sense they share value A and one-unit percentage-change approximation r. Write both models.

22. Construct linear and exponential models through the same points $(0, 1), (2, 9)$. Compare at x=1.

Challenge / HOT

23. Prove a nonconstant line and positive exponential can intersect at at most two real points.

24. Find conditions on A,m,b so $A + mx = Ab^x$ has x=0 as a solution and interpret.

AP-Style Multiple Choice

25. Constant absolute change indicates

(A) linear

(B) exponential

- **(C)** logarithmic
- **(D)** rational

26. Constant percent change indicates
- **(A)** linear
- **(B)** quadratic
- **(C)** exponential
- **(D)** constant

27. A 7% increase corresponds to factor
- **(A)** 0.07
- **(B)** 0.93
- **(C)** 1.07
- **(D)** 7

28. A 12% decrease corresponds to factor
- **(A)** 0.12
- **(B)** 0.88
- **(C)** 1.12
- **(D)** -0.12

29. For equal input steps, exponential outputs have constant
- **(A)** differences
- **(B)** ratios
- **(C)** second differences
- **(D)** sums

30. Two points determine a model only after
- **(A)** choosing a model family
- **(B)** finding a derivative
- **(C)** setting x=0
- **(D)** using a logarithm always

AP-Style Free Response

31. Two savings plans start with 1000 QAR. Plan L adds 180 QAR yearly; Plan E earns 12% yearly. (a) Write models. (b) Compare after 5 and 10 years. (c) Explain long-run behavior.

32. A population is 400 at t=2 and 1000 at t=7. (a) Build a linear model. (b) Build an exponential model. (c) Compare predictions at t=12. (d) State evidence needed to choose.

Reflection

Identify one item where a domain restriction, unit, assumption, or representation changed your reasoning. Explain in 2-3 sentences.

Original ECHS questions. Selected concepts are informed by Pearson and Holt Precalculus; no textbook exercises are reproduced.